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MEDLY

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Project Guide

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Contents

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1. Introduction

- **Problem Identified :**

Managing daily medication is challenging due to missed doses, unsafe drug combinations, and difficulty identifying symptoms. In emergencies, finding nearby pharmacies is also difficult, and no lightweight offline app currently combines all these features in one place.

- **Solution Proposed :**

Medly combines six essential health tools into one offline-first Flutter app. It enables fuzzy medicine search, dose reminders, drug interaction checks, symptom analysis, and a pharmacy finder with real-time details when online.

SDG (Sustainable Development Goals)

SDG 3:
Good Health and Well-being
Ensures better health through medicine management, reminders, drug safety, and symptom analysis

2. Objectives

- To store and manage medicine information in an organized way.
- To help users find medicines and get suggestions based on symptoms.
- To provide reminders for taking medicines and track missed doses.
- To ensure safe usage by checking drug interactions and finding nearby pharmacies.

3. Scope

1. MEDLY can be applied in the healthcare domain to help patients, caregivers, and general users manage their medications and access medicine information easily.
2. It can assist users in locating nearby pharmacies and help them track their personal medicine stock, such as how many tablets or doses they have remaining.
3. The application is also useful for doctors and healthcare workers to quickly check medicine details and drug interactions in their day-to-day practice.

4. Feature /Functionality

1. **Medicine Search & Information** Users can search for any medicine and get complete details including dosage, uses, side effects, and precautions.
2. **Medication Reminder System** Users can set reminders for their medicines with a specific time and frequency, and receive notifications even when the app is closed.
3. **Drug Interaction Checker** Users can check whether two or more medicines are safe to take together and get a severity rating with explanation for each interaction.
4. **Symptom Analyzer** Users can select their symptoms to find relevant medicines and conditions, with a reminder to always consult a doctor before taking any medication.

4. Feature /Functionality

5. Pharmacy Locator Users can find nearby pharmacies based on their current location and view the address on a map.

6. **Health Profile Management** Users can store their personal health details such as blood group, allergies, current medications, and existing medical conditions.

7. **Medicine Stock Tracking** Users can track how many tablets or doses they have remaining and receive an alert when their medicine stock is running low.

8. **Offline Mode** Users can access their medicine data, reminders, and health profile anytime without an internet connection.

5. Outcome of Project

1. Users can easily search any medicine and view its complete details like dosage, uses, and side effects.
2. Users can set medication reminders and get alerts for missed doses to never skip their medicines.
3. Users can check if their medicines are safe to take together and get warned about dangerous interactions.
4. Users can enter their symptoms and get suggestions for possible conditions and relevant medicines.
5. Users can find nearby pharmacies on a map and track how many medicines they have left.
6. Users can store and manage their complete health profile including medical history and allergies in one place.

6. Technology Stack

1. Frontend (GUI)

- Flutter (Dart) – cross-platform Android and iOS
- Material Design 3
- Provider – state management

2. API & Platform:-

- Geoapify Places API – nearby pharmacy search
- OpenFDA API – external medicine data
- Geolocator – GPS location
- HTTP – API communication

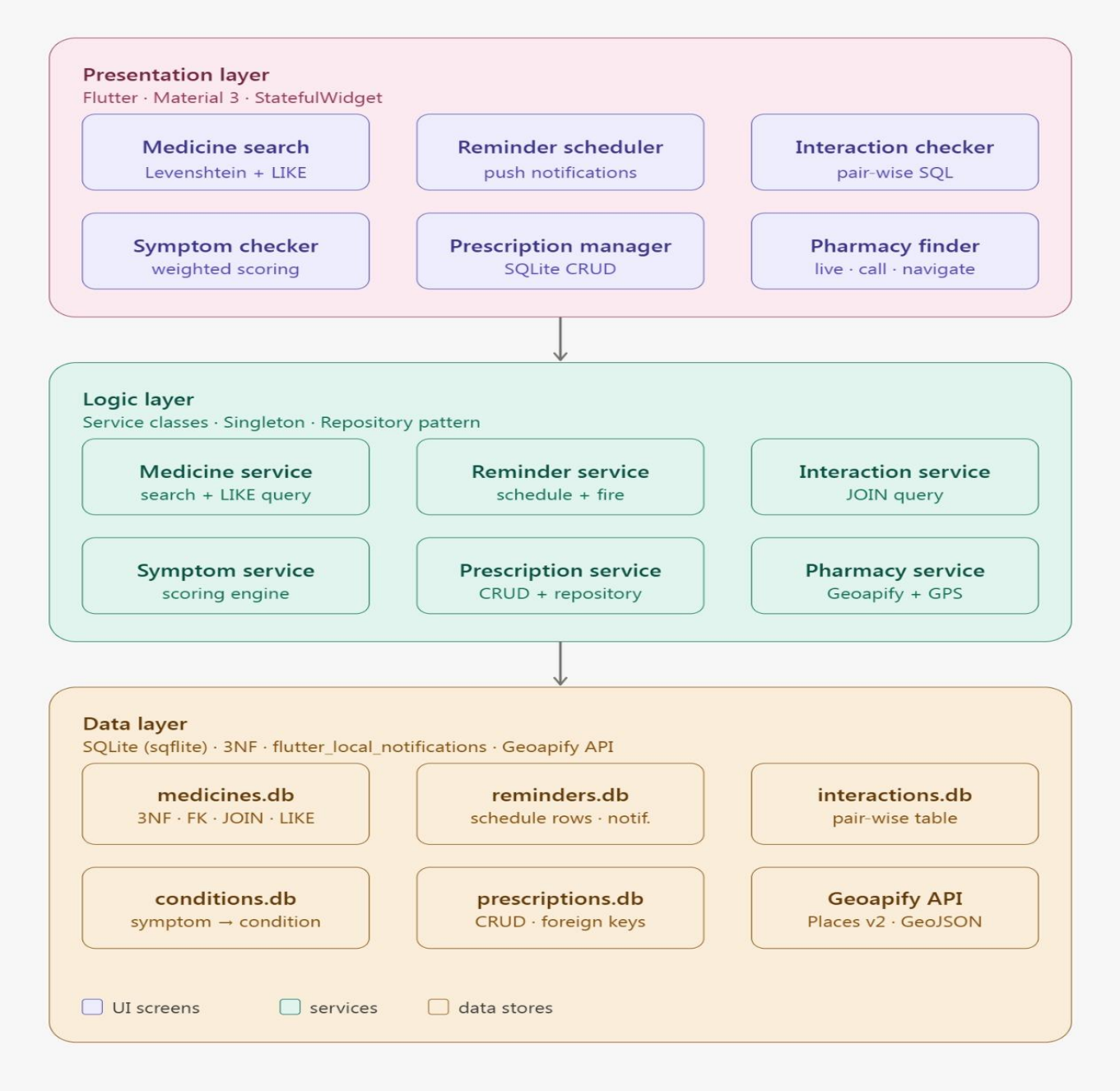
3.Database (Local Storage):-

- Supabase (PostgreSQL) – cloud database storing all app data
- Tables for medicines, reminders, drug interactions, symptoms, pharmacies, and user profiles
- Supports real-time data access and user authentication interactions, conditions

4. Back End (Logic & Services)

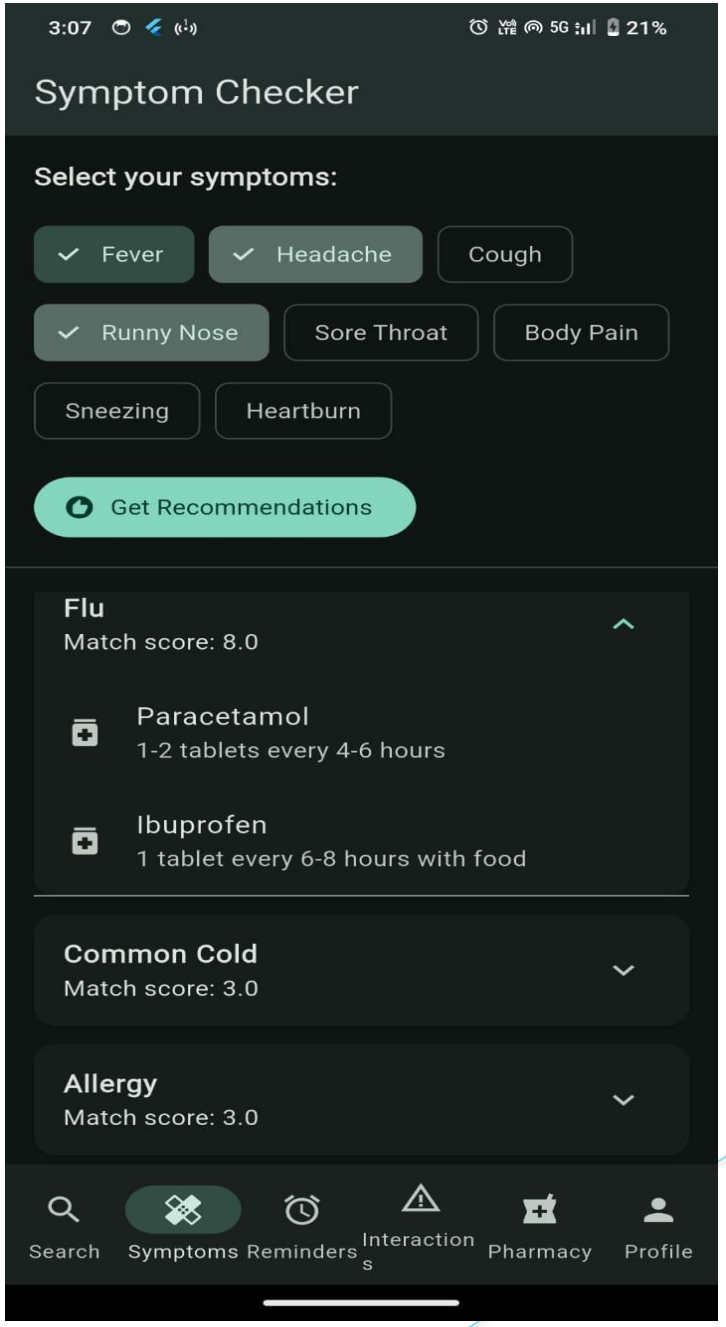
- Service layer with Singleton and Repository pattern
- Levenshtein algorithm – fuzzy medicine search
- Weighted scoring – symptom to disease mapping
- Pair-wise logic – drug interaction checking
- Periodic check every 1 minute – reminder and dose tracking

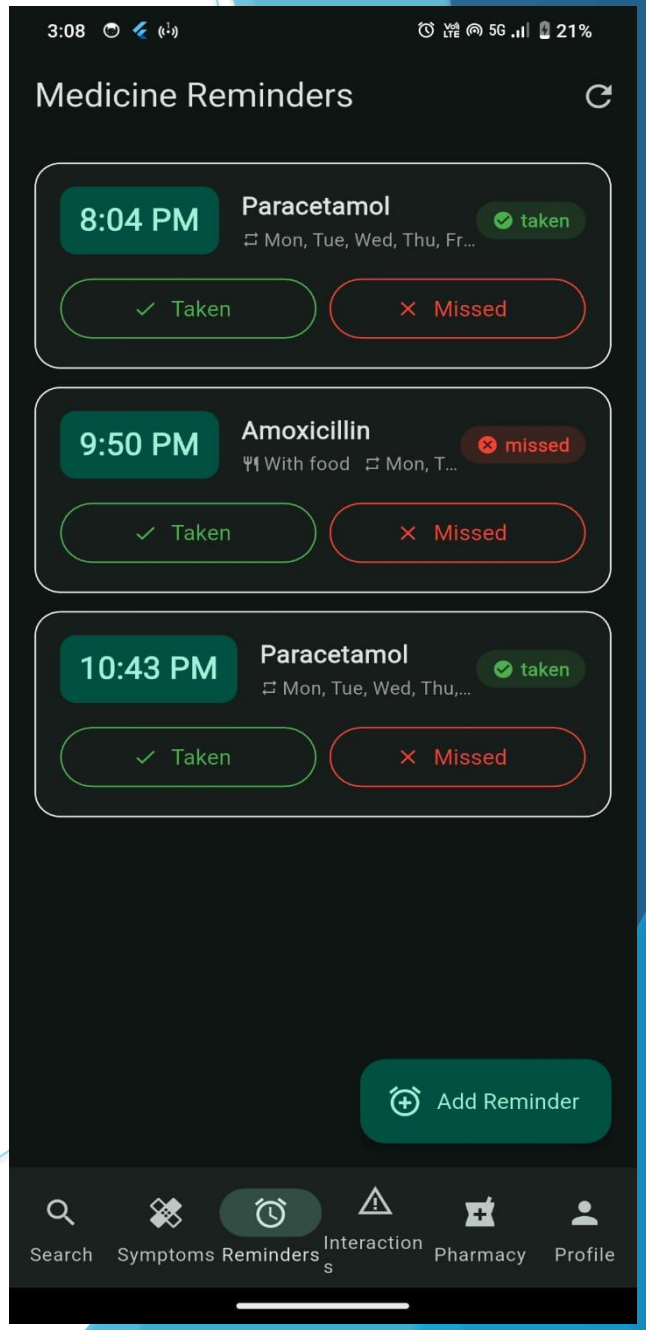
8. Block Diagram (if applicable)



10.Result

3:07
21%
Medly
Search medicines, compositions, or symptoms
Search medicines, eg. Paracetamol...
All Medicines
Paracetamol
Acetaminophen 500mg
tablet
Analgesic / Antipyretic
Ibuprofen
Ibuprofen 400mg
tablet
NSAID / Anti-inflammatory
Amoxicillin
Amoxicillin 500mg
capsule
Antibiotic (Penicillin group)
Cetirizine
Cetirizine HCl 10mg
tablet
Antihistamine
Omeprazole
Omeprazole 20mg
capsule
Proton Pump Inhibitor (PPI)
Search
Symptoms
Reminders
Interactions
Pharmacy
Profile


3:07
21%
Symptom Checker
Select your symptoms:
Fever
Headache
Cough
Runny Nose
Sore Throat
Body Pain
Sneezing
Heartburn
Get Recommendations
Flu
Match score: 8.0
Paracetamol
1-2 tablets every 4-6 hours
Ibuprofen
1 tablet every 6-8 hours with food
Common Cold
Match score: 3.0
Allergy
Match score: 3.0
Search
Symptoms
Reminders
Interactions
Pharmacy
Profile


3:08
21%
Medicine Reminders
8:04 PM
Paracetamol
taken
Mon, Tue, Wed, Thu, Fr...
Taken
Missed
9:50 PM
Amoxicillin
missed
With food
Mon, T...
Taken
Missed
10:43 PM
Paracetamol
taken
Mon, Tue, Wed, Thu,...
Taken
Missed
Add Reminder
Search
Symptoms
Reminders
Interactions
Pharmacy
Profile

3:08 5G 21%

Nearby Pharmacies

Find Real Pharmacies Near You

Lat: 19.1848, Lon: 72.9916

Search Again

4 pharmacies found nearby

Pharmacy Nearest

1.15 km

Pharmacy, 123, SC Bose Road, Jambli Naka, Thane - 400605, Maharashtra, India

Navigate

2 Vijay Medical & General Stores

2.08 km

Vijay Medical & General Stores, Sane Guruji Road, Naupada, Thane - 400601, Maharashtra, India

9757432665 | Navigate

3 BRIGHT HOMOEOPATHIC PHARMAC

2.15 km

BRIGHT HOMOEOPATHIC PHARMAC, Mavli Mandal Marg, Uthalsar, Thane - 400601, Maharas...

Search | Symptoms | Reminders | Interactions | Pharmacy | Profile

3:10 5G 21%

My Health Profile

Low Stock: Amoxicillin

2 tablets remaining — time to refill!

Basic Information

Full Name: bhumika

Age: 18

Gender: Female

Weight (kg): 35

Height (cm): 155

Blood Group: AB+

Medical Conditions

Blood Pressure

Normal | Low BP | High BP

Diabetes

None | Pre-diabeti | Type 2

+ Track Medicine

Search | Symptoms | Reminders | Interactions | Pharmacy | Profile

3:10 5G 21%

My Health Profile

Type 2

Pregnant

Allergies

Add allergy (e.g. Penicillin)

penicillin ×

Chronic Conditions

Add condition (e.g. Hypertens...

hypertension ×

Medicine Stock Tracker

Amoxicillin

+ Track Medicine

Search | Symptoms | Reminders | Interactions | Pharmacy | Profile

3:10 21%

My Health Profile

Allergies

Add allergy (e.g. Penicillin)

penicillin ×

Chronic Conditions

Add condition (e.g. Hypertens...

hypertension ×

Medicine Stock Tracker

Amoxicillin 2

2 of 10 left Exp: 2026-09-30

+ Track Medicine

Search Symptoms Reminders Interactions Pharmacy Profile

3:10 21%

My Health Profile

Allergies

Add allergy (e.g. Penicillin)

penicillin ×

Chronic Conditions

Add condition (e.g. Hypertens...

Track Medicine Stock

Medicine

Aspirin

Total Count

20

Daily Dose

1

Alert when below (tablets)

5

Expires: 2026-10-18

Add to Tracker

3:10 21%

My Health Profile

penicillin ×

Chronic Conditions

Add condition (e.g. Hypertens...

hypertension ×

Medicine Stock Tracker

Amoxicillin 2

2 of 10 left Exp: 2026-09-30

Aspirin 20

20 of 20 left Exp: 2026-10-18

+ Track Medicine

Search Symptoms Reminders Interactions Pharmacy Profile

Thank You...!!